

Research Article

Optimal model-based design of the twin-column CaptureSMB process improves capacity utilization and productivity in protein A affinity capture

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Multi-column chromatographic processes have recently been developed for protein A affinity chromatography to efficiently capture monoclonal antibodies from cell culture supernatant. In this work, the novel twin-column CaptureSMB process was compared to a batch capture process with dual loading flow rate to identify performance gains. As a case study, the isolation of a monoclonal antibody with the Amsphere JWT-203 protein A resin was investigated. Using model based optimization, both processes were optimized and compared over a wide range of operating conditions. A trade-off between productivity and capacity utilization was found, and the resulting pareto-curves showed that CaptureSMB dominates batch, except at very low productivity values. With a feed titer of 1.2 mg mL⁻¹, CaptureSMB could reach a productivity of up to 19.5 mg mL⁻¹ h⁻¹ experimentally, while maintaining relatively high capacity utilization of 63.8%. On the other hand, at maximum capacity utilization of 95.5%, a productivity of 10.2 mg mL⁻¹ h⁻¹ could be reached. This corresponds to a performance improvement with respect batch operation of about 25% in capacity utilization and 40% in productivity, for given yield and purity. CaptureSMB therefore offers a greatly increased performance over batch capture.

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1 Introduction

In chromatographic purification of biomolecules on a preparative scale, one often deals with very complex mixtures or with solutes that have very similar physical properties. This can lead to cases where baseline separation, and therefore high yield and purity, is very hard to achieve with non-affinity batch chromatography. In the case where no affinity adsorbent is available, continuous countercurrent chromatographic processes, such as SMB or MCSGP and variants thereof have been proposed

for binary, ternary and quaternary separations [1–8]. In cases where an affinity resin material is available, for example in protein A purification of monoclonal antibodies (mAbs), continuous countercurrent chromatography has not been widely used due to the high purities that could be obtained using single-column batch processes available [9, 10]. However, protein A affinity resins usually exhibit relatively strong mass transfer limitations, leading to shallow breakthrough curves, and therefore low resin utilization [11, 12]. To counteract these limitations, several different continuous multi-column setups have been proposed and implemented, such as periodic counter-current (PCC) [13, 14] and sequential multi-column chromatography (SMCC) [15, 16] processes, which use sequential loading onto three or more columns to achieve increased performance. Recently, a continuous, two-column capture process has been proposed, which allows for higher productivity and higher capacity utilization

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Abbreviations: CIP, cleaning in place; mAb, monoclonal antibody

tion than comparable batch processes, while retaining comparable separation performance in terms of yield and purity [17]. Higher productivity means that a larger amount of product can be processed per unit time and unit volume of resin, which reduces the time required to process a given amount of feed stock, while higher capacity utilization means that more product can be processed per unit volume of resin and per cycle, which reduces the number of load, elution and regeneration cycles needed to process a given amount of product. Therefore, capacity utilization means that more product can be produced per unit volume of resin within the life time of the resin. Replacing the resin is a major component of the overall production cost, especially in cases where the resin is expensive [18].

When dealing with a novel process, gaining process understanding is paramount to design an efficient and robust process and to determine the operating regions and optimal process conditions. In order to avoid large time-consuming and resource-consuming experimental campaigns, especially in cases like protein A chromatography for monoclonal antibodies, where both the column material and the feed are usually very expensive, process modeling and simulation has long since been a vital tool in gaining this understanding [19–22]. Many different models are available to predict the behavior of chromatographic systems under preparative conditions, such as the ideal model [23], Langmuir type models [24, 25], or the steric mass action model [26]. However, these models are generally not very well equipped to predict the behavior of systems with high mass transfer resistances, which are usually observed in protein chromatography. In such scenarios, pore diffusion or general rate models have been applied [20, 27, 28]. In the case of a large protein, such as a mAb, these models can be augmented with features that account for hindrance of mass transfer by proteins blocking pores, which has been shown recently [29, 30]. However, when optimizing a multi-column chromatographic process, these elaborate models tend to be too computationally expensive to complete calculations in a reasonable time frame; therefore a relatively simple model combined with semi-empirical correlations is usually preferable in these cases [31, 32]. In this work, such a model for the twin-column CaptureSMB process is developed, validated experimentally and used to find optimal operating points for the process. The model is held general enough to be easily applied to batch chromatography and other multi-column processes as well.

2 Process descriptions

In order to have a fair comparison between the processes, different numbers of resin volumes are used in the CaptureSMB and the batch process. In the batch process, one column is either the same bed height as one column in the

CaptureSMB process, or the column in the batch process is twice the bed height of one column in the CaptureSMB process, leading to the same total bed height.

In addition, the purity is assumed to be the same in both processes, and the yield is kept constant at a high value by enforcing a constraint on the breakthrough (see below). Product quality is assumed to be comparable between the two processes and independent on the operating parameters, which has been proven previously in [17]. Since the recovery and regeneration step can be optimized independently, it is assumed to be fixed and equal in both processes and is not part of the comparison.

2.1 Batch chromatography with dual loading flow rate

The affinity batch chromatography process uses a single column, as schematically depicted in Fig. 1A. In the batch process, the column is first equilibrated with a buffer that allows binding of the target compound to the resin material in the column. When the column is equilibrated, the feed solution containing the target compound is loaded onto the column, where the target compound is retained while non-binding impurities are flowing through, ending up in the waste stream (W). One or more washing steps, where the column is flushed with different buffers, can then be applied to clear more impurities. After these washing steps, the target compound is recovered by applying a buffer that allows desorption of the compound from the resin. After the product has been collected (P), the column is cleaned in place (CIP) with a CIP-solution to remove any strongly binding impurities and to sanitize the equipment. When the CIP step is finished, the column is re-equilibrated, preparing the column for the next load-elute cycle.

In the batch process considered, the loading step is further divided into sub-steps. The loading phase encompasses two sub-steps with loading at two different flow rates, Q_{PL} and Q_L for durations t_{PL} and t_L , respectively. The indices PL and L stand for preload and load, respectively.

This dual loading flow rate strategy has been proposed for improvement of batch capture performance [9]. It is used in this work in order to ensure a fair comparison between batch chromatography and CaptureSMB, which also uses a dual loading flow rate strategy.

The entire elution phase including wash, elute, CIP and re-equilibration has a total duration denoted by t_{RR} . This leads to four design variables that completely describe the process: t_{PL} , t_L , Q_{PL} and Q_L . These are all free variables and there are therefore four degrees of freedom for optimization. As mentioned above, the optimization is only applied to the loading step, since optimal procedures for the remaining steps can be determined independently of the loading step.

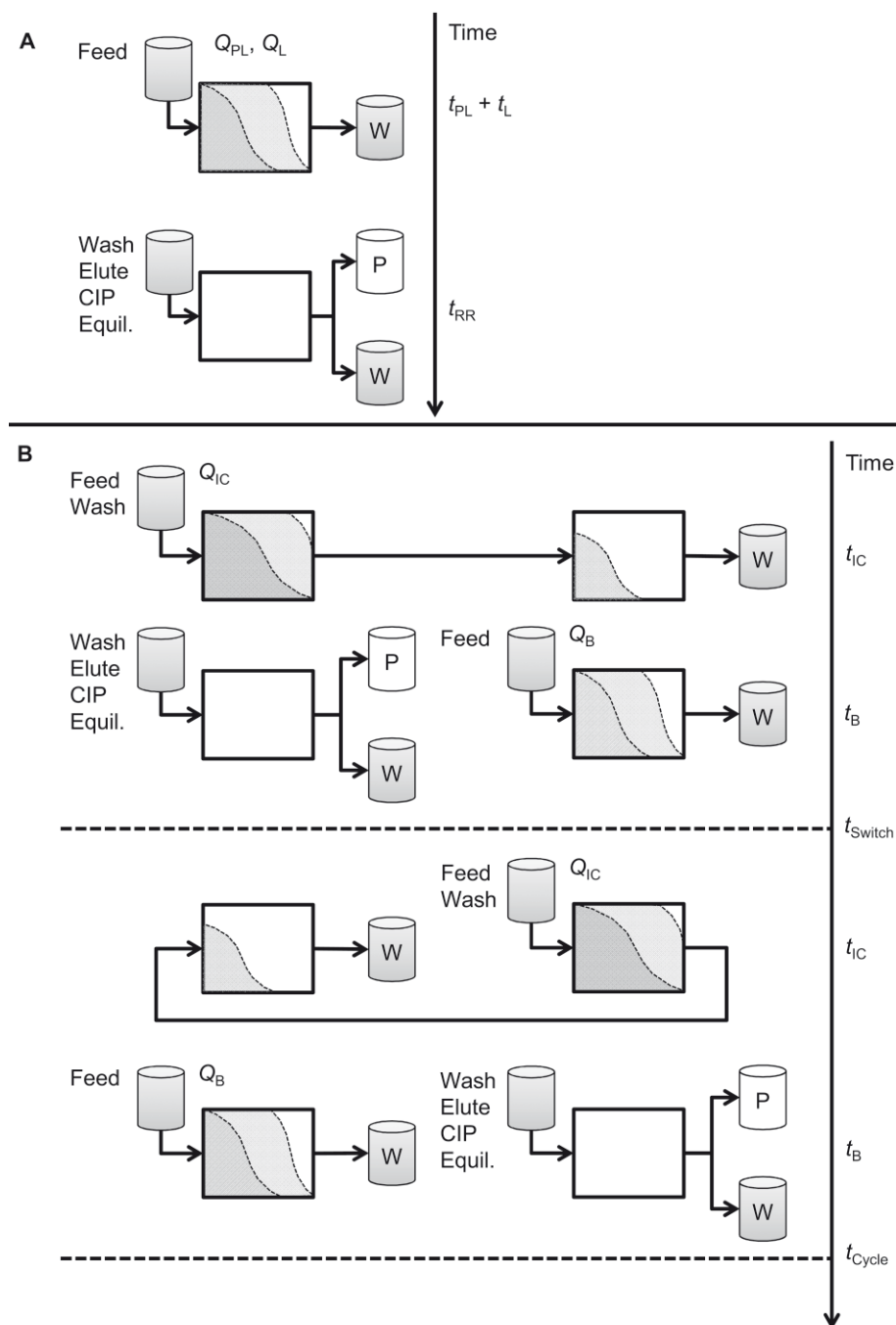


Figure 1. Process flow charts for the batch capture process (A) and the CaptureSMB process (B).

2.2 CaptureSMB

In the CaptureSMB process, two identical columns are used to recover the target compound from the feed mixture. After the columns have been equilibrated, they are interconnected by connecting the outlet of the first column to the inlet of the second column. For a certain time, the feed solution is loaded onto the first column. The fact that there is a second column behind the first one allows for longer feeding compared to the batch case,

because the eventual breakthrough is caught by the second column instead of being lost. After this time, the first column enters the recovery and regeneration step, which is equivalent to the batch process described in 3.1. During this step, the now disconnected second column is further loaded with feed. As soon as the first column is re-equilibrated and ready to be loaded again, the columns swap roles and continue with the next interconnected step. A process schematic is shown in Fig. 1B. In the first part of the switch, the interconnected phase (IC, top), the

columns are connected in series and feed is loaded with a flow rate Q_{IC} for a time t_{IC} . At the end of the interconnected phase, the wash step starts for the fully loaded column, while the columns are still interconnected, to catch any product that elutes during this early wash step. Non-adsorbing impurities flow through and go to waste (W). Next, the batch phase (B) follows for a duration t_B , where the fully loaded column is washed further and more impurities are cleared. After the washing steps, the pure product (P) is eluted and collected. CIP and equilibration steps follow to prepare the column for the next switch. In the meantime, the other column is loaded further with a feed flow rate Q_B . After the first switch, at $t_{Switch} = t_{IC} + t_B$, before any product in the second column starts breaking through, the columns switch roles and go through the same procedure again with reversed roles (lower part). After one cycle, at $t_{Cycle} = 2(t_{IC} + t_B)$, the columns switch roles again and are back in the initial configuration.

The CaptureSMB process has four design variables that completely describe the process: t_B , t_{IC} , Q_B and Q_{IC} , which are the duration of the batch and interconnected phases, and the loading flow rates during the batch and interconnected phases, respectively. Since one of the columns undergoes recovery and regeneration during the batch phase, this procedure fixes the batch time t_B . This leaves three degrees of freedom to optimize the process.

3 Materials and methods

3.1 Monoclonal antibody

The cell culture supernatant used in the experimental parts of this study was obtained from JSR (Tokyo, Japan) and contained a monoclonal IgG₁-type antibody in a concentration of $c_{Feed} = 1.2 \text{ mg mL}^{-1}$. Higher feed concentrations were achieved by spiking the feed with previously captured mAb from the same feed.

3.2 Stationary phase and buffers

The protein A stationary phase Amsphere JWT-203 (JSR, Tokyo, Japan) was used for all experiments, in pre-packed columns of dimensions $0.5 \times 5 \text{ cm}$ (Atoll, Weingarten, Germany). Two or four such columns were joined in series to form new columns with bed heights of 10 and 20 cm. The buffers were: 20 mM phosphate, 150 mM NaCl, pH 7.5 as equilibration, washing and binding buffer (A); 20 mM phosphate, 1 M NaCl, pH 7.5 as second washing buffer (B); 20 mM citrate pH 3.2 as elution buffer (C); and 0.1 M NaOH as cleaning-in-place (CIP) solution.

3.3 Preparative chromatography operating conditions

All preparative batch and CaptureSMB runs were conducted on Contichrom lab-scale equipment (ChromaCon AG, Zurich, Switzerland). The following protocol was used for all elution phases, after a column has been fully loaded:

- Washing: five column volumes buffer A, five column volumes buffer B, five column volumes buffer A
- Elution: five column volumes buffer C
- CIP: 15 min contact time with the CIP solution
- Equilibration: five column volumes buffer A

The very first equilibration step needed before starting the processes is negligible due to the cyclic nature of the processes and the high number of switches to be run. The recovery and regeneration step was carried out at $Q_{RR} = Q_{IC}$.

3.4 Analytics

The concentration of mAb in all samples was determined using the method described in [17] using a Poros® A-20 analytical protein A chromatography column ($2.1 \times 30 \text{ mm}$), obtained from Life Technologies (Carlsbad, CA, USA) on an Agilent 1100 series HPLC (Santa Clara, CA, USA). The mobile phases used were 20 mM phosphate, 150 mM NaCl, pH 7.5 (binding and washing buffer), 10 mM HCl, 150 mM NaCl, pH 2.0 (elution buffer) and 20% acetic acid (regeneration solution) and the flow rate was 1.0 mL min^{-1} .

4 Process model

4.1 Model equations with empirical mass transfer model

The aforementioned chromatographic processes are modeled using a lumped kinetic model [19]:

$$\begin{aligned} \frac{\partial c}{\partial t} &= -v \frac{\partial c}{\partial x} + D_L \frac{\partial^2 c}{\partial x^2} - \phi \frac{\partial q}{\partial t} \quad t \in [0, t_{Switch}], x \in [0, L_{Col}] \\ \frac{\partial q}{\partial t} &= k_m (q^* - q) \end{aligned} \quad (1)$$

where c is the liquid phase concentration of the protein, t denotes the time, $v = Q/(A_{Col}\epsilon)$ is the interstitial velocity, where Q is the volumetric flow rate, A_{Col} is the column cross sectional area and ϵ is the bed porosity, x is the coordinate along the longitudinal axis of the column, L_{Col} is the length of one column, D_L is the apparent axial dispersion coefficient, $\phi = (1 - \epsilon)/\epsilon$ is the phase ratio of the column, q is the solid phase concentration of the protein, k_m is the mass transfer coefficient and q^* is the equilibrium solid phase concentration of the protein.

These partial differential equations are subject to the following boundary and initial conditions:

$$\begin{aligned} c(t=0, x) &= c_0(x) \\ q(t=0, x) &= q_0(x) \\ c(t, x=0) &= c_{in}(t) + \frac{D_L}{v} \frac{\partial c}{\partial x} \Big|_{x=0} \\ \frac{\partial c}{\partial x} \Big|_{x=L_{Col}} &= 0 \end{aligned} \quad (2)$$

Note that the inlet concentration $c_{in}(t)$ changes with time, because the columns undergo different operations (breakthrough uptake, load, wash, elute, CIP, equilibrate).

The apparent axial dispersion coefficient D_L can be estimated from the reduced van Deemter equation with the correlation [27]:

$$D_L = A \frac{d_p}{2} v \quad (3)$$

where A is the intercept of the reduced van Deemter equation and d_p is the average particle diameter.

The bed porosity ε can be measured using a pulse injection of a non-adsorbing species.

For the mass transfer coefficient k_m , an empirical correlation is used that approximates the effect of hindered mass transfer due to pore blockage and other effects [32]:

$$k_m = k_m^{\max} \left(S_1 + (1 - S_1) \left(1 - \frac{q}{q_{sat}} \right)^{S_2} \right) \quad (4)$$

where $k_m^{\max}$ is the maximum mass transfer coefficient when no protein is adsorbed, S_1 is a coefficient describing the maximum hindrance, for which $0 < S_1 \leq 1$, the coefficient $S_2 > 0$ models the non-linearity of the increase in hindrance and q_{sat} is the saturation capacity of the resin. The maximum mass transfer coefficient and the parameters of the correlation are fitted from breakthrough curves at different flow rates and concentrations.

The adsorption of the protein is described using a Langmuir isotherm with the pH value as the modifier [2]:

$$\begin{aligned} H &= H_{Ref} \left(\frac{pH}{pH_{Ref}} \right)^N \\ q^* &= \frac{Hc}{1 + \frac{Hc}{q_{sat}}} \end{aligned} \quad (5)$$

where H is the Henry coefficient (which is proportional to the retention factor $k_{0,i}$), H_{Ref} is the Henry coefficient measured at the reference pH value, pH_{Ref} is the reference pH value and N describes the change of the Henry coefficient with the pH value. H_{Ref} and q_{sat} are fitted from breakthrough curves at different flow rates and concentrations, and N is fitted from elution experiments.

The resulting partial differential equations were discretized along the coordinate using a first order central finite differences method, and the resulting system of ordinary differential equations was solved in FORTRAN using the solver DLSODES from ODEPACK [33] (Hindmarsh, A. C., Netlib ODEPACK, <http://www.netlib.org/odepack>, retrieved 06/2013). The number of grid points was fixed at 100 per column, since a further increase did not show any changes in the solution behavior. The column switching in the CaptureSMB process was conducted numerically by swapping the internal profiles of the columns. This way, the same boundary conditions could be used in every switch. The model is suitable to simulate batch experiments and the CaptureSMB process, but is also kept general enough to be adapted to any multi-column process.

4.2 Fixed operating parameters

Some parameters are not part of the optimization, namely the column dimensions and the recovery and regeneration protocol. The following column dimensions were used in the simulations: $A_{Col} = 0.20 \text{ cm}^2$ and $L_{Col} = 5$ or 10 cm for a single column in the CaptureSMB case, and $L_{Col} = 5, 10$ or 20 cm for the column in the batch case.

The elution protocol mentioned above (see 4.3) fixes the recovery and regeneration time (which, in the case of CaptureSMB, is equal to the batch loading time) as follows:

$$t_{RR} = t_W + t_{Elu} + t_{CIP} + t_{Equil} \quad (6)$$

where W stands for wash, Elu stands for elution, CIP stands for cleaning in place and $Equil$ stands for re-equilibration.

4.3 Process constraints for yield, purity and pressure drop

As mentioned above, the comparison discussed below is performed with constant purity and yield. The impurity clearance is largely dependent on the type of stationary phase and the washing procedure. These are kept identical in the two processes, and it is assumed that the parameters chosen fulfil any purity requirements; therefore the purity is not modelled. On the other hand, yield is a strong function of the operating parameters such as flow rates and durations of the different process steps and it is properly calculated by the model. However, for a high value product such as monoclonal antibodies, a process is not attractive if too much product is lost; yield is therefore treated as a constraint rather than an optimization variable. In the batch process, maximum yield is ensured by loading only to a volume that corresponds to a certain breakthrough value (e.g. 90% of the volume corresponding to 1% breakthrough [10]). This safety factor of 90%

is rather arbitrary; therefore it has not been considered in the optimizations and the constraint is simply that the breakthrough after loading should never exceed 1%. Similarly for the CaptureSMB process, the breakthrough of the second column, which has no column connected to its outlet, must not exceed 1%. With these constraints it is assured that in the comparison discussed below, all process runs lead to equivalent yield values.

In all cases, the pressure drop over the columns must not exceed the maximum allowed pressure drop. The pressure drop is estimated using the following correlation [11]:

$$\Delta P = \gamma u L \quad (7)$$

where γ is a coefficient, $u = Q/A_{\text{Col}}$ is the superficial velocity and L is the total bed height, i.e. number of interconnected columns times L_{Col} .

4.4 Process optimization

The aim of the process optimization was to maximize the productivity or the capacity utilization. All optimization tasks were done with the built-in genetic algorithm function GA of MATLAB (Mathworks), which implements an evolutionary procedure that drives a randomly distributed initial population of candidate solutions, in this case four sub-populations of 20 members each, towards functional minima. By choosing a relatively large number of individuals, coverage of the solution domain can be improved and the possibility of finding the global optimum is increased.

The productivity represents the amount of antibody which can be produced per unit time and unit resin volume. For the batch process, it can be written as follows

$$P_{\text{Batch}} = Y \cdot \frac{c_{\text{Feed}}(t_{\text{PL}}Q_{\text{PL}} + t_{\text{L}}Q_{\text{L}})}{V_{\text{Col}}(t_{\text{PL}} + t_{\text{L}} + t_{\text{RR}})} \quad (8)$$

where Y is the process yield, c_{Feed} is the mAb concentration in the feed and $V_{\text{Col}} = A_{\text{Col}} \cdot L_{\text{Col}}$ is the volume of one column.

With the constraints discussed above, this leads to the following optimization problem

$$\begin{aligned} &\text{maximize}_{t_{\text{PL}}, t_{\text{L}}, Q_{\text{PL}}, Q_{\text{L}}} P_{\text{Batch}}(t_{\text{PL}}, t_{\text{L}}, Q_{\text{PL}}, Q_{\text{L}}) \\ &\text{subject to} \quad c(t = t_{\text{PL}} + t_{\text{L}}, z = L_{\text{Col}}) \leq 0.01 c_{\text{Feed}} \\ &\quad (\gamma u L)_{\text{max}} \leq \Delta P_{\text{max}} \\ &\quad 0 < t_{\text{PL}}, 0 \leq t_{\text{L}}, 0 < Q_{\text{PL}}, 0 < Q_{\text{L}} \end{aligned} \quad (9)$$

Note that the first constraint $c(t = t_{\text{PL}} + t_{\text{L}}, z = L_{\text{Col}}) \leq 0.01 c_{\text{Feed}}$ is equivalent to a yield constraint and assures that the yield is kept high and constant.

A similar expression can be derived for the CaptureSMB process

$$P_{\text{CSMB}} = Y \cdot \frac{c_{\text{Feed}}(t_{\text{IC}}Q_{\text{IC}} + t_{\text{B}}Q_{\text{B}})}{2V_{\text{Col}}(t_{\text{IC}} + t_{\text{B}})} \quad (10)$$

where $V_{\text{Col}} = A_{\text{Col}} \cdot L_{\text{Col}}$ is the volume of one column in CaptureSMB. The expression (10) can be used in (9) with analogous constraints and optimization variables.

As mentioned before, with relatively expensive and short-lived resins, maximizing the capacity utilization becomes a very important objective. The capacity utilization is equivalent to the amount of mAb produced per resin volume and cycle normalized with the equilibrium binding capacity at feed concentration. It can be calculated as

$$CU_{\text{Batch}} = Y \cdot \frac{c_{\text{Feed}}(t_{\text{PL}}Q_{\text{PL}} + t_{\text{L}}Q_{\text{L}})}{V_{\text{Col}}(1 - \varepsilon)q_{\text{Feed}}^*} \quad (11)$$

where CU stands for the capacity utilization and q_{Feed}^* is the equilibrium binding capacity at feed concentration which can be calculated using Eq. (5). Using the same constraints as in Eq. (9), an analogous optimization problem can be formulated, with CU as the objective to maximize. Similarly, in CaptureSMB, the capacity utilization is defined as:

$$CU_{\text{CSMB}} = Y \cdot \frac{c_{\text{Feed}}(t_{\text{IC}}Q_{\text{IC}} + t_{\text{B}}Q_{\text{B}})}{V_{\text{Col}}(1 - \varepsilon)q_{\text{Feed}}^*} \quad (12)$$

Note that the factor 2 in the denominator cancels compared to Eq. (10), since in order to elute both columns, an entire cycle, i.e. two switches, is needed, so Eq. (10) must be multiplied with $2(t_{\text{IC}} + t_{\text{B}})$ before normalizing with the equilibrium binding capacity at feed concentration.

The presence of two competing objectives, in this case productivity and capacity utilization, can lead to a trade-off scenario where one of the objectives cannot be improved without having a negative impact on the other. In this case, a set of pareto-optimal operating points can be found, for which no other feasible operating points exist that are better in both objectives at the same time. To find a set of pareto-optimal points, the optimization problem in Eq. (9), using Eq. (11) and (12) as optimization goal, is modified by adding the constraint:

$$P_{\text{Target}} \leq P \quad (13)$$

where P_{Target} , the target productivity, is varied between the optimum values found from the optimization problem in Eq. (9) with productivity as objective, and the productivity value found by optimizing Eq. (9) for maximum capacity utilization.

5 Results

5.1 Model fitting

The reduced plate height of pulse injections with a non-adsorbing solute at different flow rates was determined using the method of moments [34]. Linear regression of the data yields the parameter $A = 35.13 \pm 5.451$ which is used in Eq. (3) to calculate the apparent axial dispersion coefficient.

From breakthrough experiments at different flow rates and feed concentrations, the parameters describing the adsorption behavior, $k_m^{\max}$, S_1 , S_2 , H_{ref} and q_{sat} can be estimated. This is done by minimizing the sum of square errors using a genetic algorithm. In total, nine breakthrough curves at five different flow rates between 0.25 and 3 mL min⁻¹ and three different feed concentrations between 1.2 and 4.6 mg mL⁻¹ were used to fit the model parameters. The relatively low value of the maximum mass transfer coefficient $k_m^{\max} = 0.1800 \text{ min}^{-1}$, obtained by fitting, indicates a strongly hindered mass transfer. Furthermore the results indicate that the mass transfer is expected to decrease by almost 38% at saturation, as indicated by the value found for $S_1 = 0.6245 \pm 1.739$ (Eq. 4) in order to fit the asymmetry and the broadness of the breakthrough curves. A list of all parameters used can be found in the Supporting information, Table S1.

Examples of the fitted breakthrough curves are shown in Fig. 2. It is clear that the model can replicate the elution behavior over a wide range of different flow rates and different feed concentrations. An example of simulated and experimental UV signals can be found in the Supporting information, Fig. S1.

A pressure drop coefficient of $\gamma = 0.01020 \pm 5.436 \times 10^{-4} \text{ bar min cm}^{-2}$ used in Eq. (7) was determined by linearly fitting the pressure drop values measured at different flow rates.

In order to assess whether the model is able to reasonably reproduce the CaptureSMB process, different operating points were simulated and the output compared to experimental results at these points. The points included a point at high productivity (run 1), a point at high capacity utilization (run 3) and an intermediate point (run 2). Additionally, runs 4 and 5 were used as verification runs, where the operating parameters have been taken from the pareto-optimal set of operating points (see 6.4) and then run experimentally. All corresponding operating parameters are reported in the Supporting information, Table S2. To ensure feasible operation, a safety factor has been added to the operating parameters, i.e. 20% reduced batch loading flow rate. The experiments were run for three cycles at each point which was sufficient to reach cyclic steady state.

From the data shown in Table 1, it can be seen that the model is able to reproduce the experiments with good accuracy over a wide range of flow rates and load-

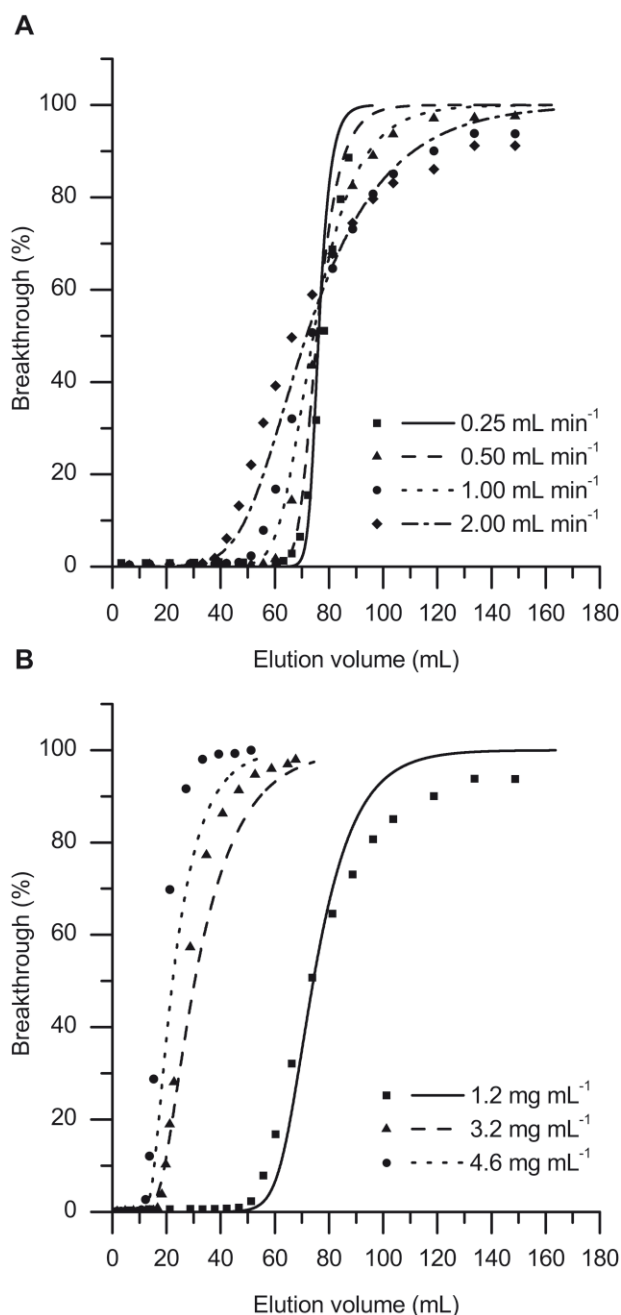


Figure 2. (A) Breakthrough curves at different flow rates (filled symbols) with model data (lines). The feed concentration was 1.2 mg mL⁻¹ in these cases. (B) Breakthrough curves at different feed concentrations (filled symbols) with model data (lines). The feed flow rate was 1.0 mL min⁻¹ for the experiments with feed concentrations of 1.2 and 4.6 mg mL⁻¹, and 1.5 mL min⁻¹ for the run with 3.2 mg mL⁻¹.

ing times. The relative error was smaller than 10% in all performance measures considered, except for the outlet concentration of run 2. Note that all yield losses in both the experiments and the model results are due to losses in the wash, not due to breakthrough. It appears that due

Table 1. Comparison of model predictions with experimental data for the performance of the CaptureSMB process in different conditions. In parentheses: Relative errors of the simulation output compared to the experimental data

Run #	Yield Y (%)		Mass balance (%)		Productivity P (mg mL ⁻¹ h ⁻¹)		Capacity utilization CU_{feed} (%)		Outlet concentration c_{out} (mg mL ⁻¹)	
	Exp.	Sim.	Exp.	Sim.	Exp.	Sim.	Exp.	Sim.	Exp.	Sim.
1	92.9	93.9 (+1.1%)	102.2	100.0	17.7	18.3 (+3.4%)	67.5	65.3 (-3.3%)	5.48	5.93 (+8.2%)
2	85.3	92.7 (+8.7%)	93.7	100.0	15.7	17.1 (+8.9%)	82.9	76.2 (-8.1%)	6.06	6.91 (+14%)
3	88.1	88.6 (+0.6%)	97.2	100.0	10.2	10.4 (+2.0%)	95.5	96.0 (+0.5%)	6.9	7.39 (+7.1%)
4	93.6	94.5 (+1.0%)	101.7	100.0	19.5	21.0 (+7.7%)	63.8	61.1 (-4.2%)	5.07	5.55 (+9.5%)
5	90.2	94.0 (+4.2%)	99.7	100.0	16.0	16.9 (+5.6%)	74.8	72.4 (-3.2%)	6.1	6.57 (+7.7%)

to this, increasing the load has a slightly negative impact on yield, but there is no significant correlation. The fact that the capacity utilization seems to be underestimated while the outlet concentration is overestimated might be an indication that the model underestimates the steady state preload, i.e. what fraction of the feed loaded in the interconnected phase breaks through and is loaded onto the second column. If this preload is underestimated, more product ends up in the first column before recovery, leading to a higher concentration, while less product is internally recycled, leading to less product being in the process overall and therefore lower capacity utilization. However, the mass balance not closing in the experiments certainly also plays a role in the inaccuracies in model prediction. As can be seen, run 4, which is optimized for productivity, offers higher productivity than what was reached in previous experiments that were designed without prior model based optimization. Since run 3 was already very close to the global optimum in terms of capacity utilization, the last point (run 5) was taken from the middle of the pareto curve. Also in this intermittent region, the model is able to replicate the behavior of the system.

5.2 Internal concentration profiles

One might expect that using additional columns for sequential could further increase the load and therefore the capacity utilization, but considering the internal concentration profile just before recovery and regeneration that results from solving Eq. (1) and (2) at the operating point with maximum capacity utilization (solid line in Fig. 3), it is obvious that only a very small fraction of the capacity is still available, which is represented as the small grey area in the top right corner of the dashed rectangle, even though the profile is contained in only two

columns. Therefore, we can conclude that two interconnected columns are sufficient for an efficient continuous capture process. The internal concentration profile at maximum productivity (dotted line in Fig. 3) is shown as a reference, which shows a lower capacity utilization, as expected. At the interface of the two columns (vertical dashed line at 10 cm), a discontinuity is visible; This is an effect of the Danckwert's boundary conditions (Eq. 2).

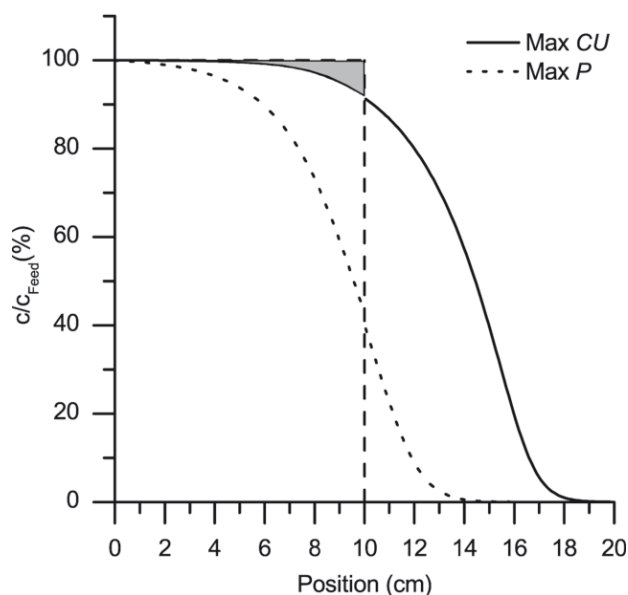


Figure 3. Internal concentration profile in the liquid phase of the CaptureSMB process at the end of the interconnected phase, when the first column is fully loaded. The two profiles shown correspond to a process run at maximum capacity utilization (solid line) and an operating point at maximum productivity (dashed line). The grey area represents the resin capacity not utilized at maximum capacity utilization.

Table 2. Summary of optimized operating points (single objective) with operating parameters

Variable	Symbol	Units	Value			
Optimization Goal	–	–	Productivity		Capacity Utilization	
Process	–	–	Batch	CSMB	Batch	CSMB
Productivity	P	$\text{mg mL}^{-1} \text{ h}^{-1}$	19.7	27.1	1.80	16.8
Capacity utilization at c_{Feed}	CU_{Feed}	%	71.2	79.1	99.2	94.8
Load	–	mg mL^{-1}	32.3	35.9	45.0	43.0
Outlet concentration	c_{out}	mg mL^{-1}	6.46	7.18	9.00	8.60
Buffer consumption	BC	mL mg^{-1}	0.813	0.766	0.584	0.640
Preload / Interconnected time	$t_{\text{PL}}, t_{\text{IC}}$	min	42.56	5	699	40.5
Preload / Interconnected flow rate	$Q_{\text{PL}}, Q_{\text{IC}}$	mL min^{-1}	2.5	2.5	0.105	1.40
Load time	t_{L}	min	1.58	N / A	752	N / A
Load / Batch flow rate	$Q_{\text{L}}, Q_{\text{B}}$	mL min^{-1}	0.26	1.45	0.1	0.45
Cycle time	t_{cycle}	min	98.4	79.4	1496	154

5.3 Process optimization

A summary of optimum operating points for single objective optimization and their respective values is reported in Table 2 for both processes considered. In this case, the single column length for CaptureSMB was 10 cm, while the batch process was simulated with the same total bed height, resulting in a column length of 20 cm. The feed concentration was 1.2 mg mL^{-1} in all cases.

When maximizing the productivity, the batch process reached a value of $19.7 \text{ mg mL}^{-1} \text{ h}^{-1}$, while the CaptureSMB process reached $27.1 \text{ mg mL}^{-1} \text{ h}^{-1}$. It is clear that the CaptureSMB process shows better performance with respect to productivity, and in addition has a better capacity utilization (79.1 compared to 71.2%) and therefore higher load (35.9 mg mL^{-1} compared to 32.3 mg mL^{-1}) and a higher outlet concentration (7.18 mg mL^{-1} compared to 6.46 mg mL^{-1}). This also means that fewer cycles are required to process an equal amount of antibody, which leads to slower resin deterioration and therefore lower resin costs.

When maximizing the capacity utilization on the other hand, the batch process shows slightly better values, reaching a capacity utilization of 99.2%, resulting in a load of 45.0 mg mL^{-1} and an outlet concentration of 9.00 mg mL^{-1} , compared to a capacity utilization of 94.8% in the case of CaptureSMB, leading to a load of 43.0 mg mL^{-1} and an outlet concentration of 8.60 mg mL^{-1} . However, in terms of productivity at the operating point with maximum capacity utilization, CaptureSMB shows a vastly superior performance, yielding $16.8 \text{ mg mL}^{-1} \text{ h}^{-1}$ compared to only $1.80 \text{ mg mL}^{-1} \text{ h}^{-1}$ in the batch case. The difference in capacity utilization is due to the fact that in order to run CaptureSMB, a preload is required that occupies a fraction of the capacity in steady state operation, effectively decreasing the amount of new product that can be loaded per cycle, which is the basis for the calculation of the capacity utilization. In all cases, the specific buffer consumption is comparable

between the two processes. At maximum productivity, the CaptureSMB process shows slightly lower buffer consumption (0.766 mL mg^{-1} compared to 0.813 mL mg^{-1}), while at maximum capacity utilization, the CaptureSMB process utilizes slightly more buffer (0.640 mL mg^{-1} compared to 0.584 mL mg^{-1}).

5.4 Pareto optimal operating points and influence of column length

It becomes apparent that there is a trade-off between maximizing capacity utilization (i.e. minimizing resin consumption) and maximizing productivity (i.e. speeding up the process). Using the algorithms detailed in 5.4, it is possible to generate pareto-curves, on which all points are equally optimal when taking into account both objectives. These curves allow for the complete and exhaustive comparison between the two processes.

A set of pareto-optimal points in the capacity utilization/productivity plane are shown in Fig. 4. In particular, we consider the batch process with column lengths of 5, 10 and 20 cm and CaptureSMB processes with single column lengths of 5 and 10 cm, at a constant feed concentration of 1.2 mg mL^{-1} . This allows us to estimate the effect of column length on the performance of the processes. In all cases, the pareto curves of different column lengths cross, which means that with short columns, a higher maximum productivity can be achieved, while with long columns it is easier to achieve higher capacity utilization.

It is clear that, except at very low productivity, CaptureSMB always dominates batch processes with the same or higher column lengths and is therefore an improvement in terms of both capacity utilization and productivity. Additionally, CaptureSMB enables operating points with productivity values far above the maximum possible productivity in batch mode, while simultaneously maintaining high capacity utilization and therefore increasing the load and the outlet concentration.

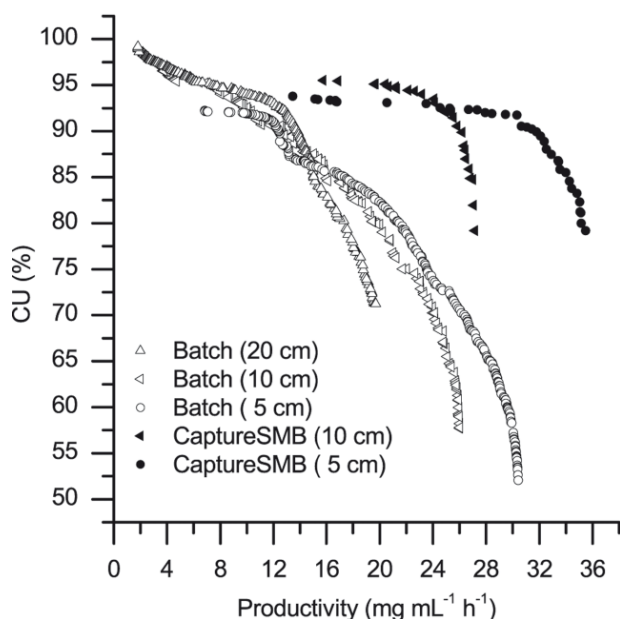


Figure 4. Pareto-optimal operating points for the batch process (open symbols) and the CaptureSMB process (full symbols) with different column lengths. In the CaptureSMB case, the column length refers to the length of a single column.

It can be seen that in the batch case, it is always possible to run the process at such low flow rates that the capacity utilization tends to 100%, regardless of column length, but this leads to very low productivity values. In the CaptureSMB process, a preload is needed; therefore there is a difference in maximum capacity utilization between the different column lengths, albeit very small. So in cases where very low productivity, i.e. almost 10 times lower than the maximum achievable productivity, is acceptable, and very high capacity utilization, i.e. close to 100%, is needed, the batch process is preferred over the CaptureSMB process.

When examining the change of the operating parameters t_{IC} , Q_{IC} and Q_B along the pareto front of the CaptureSMB process, one finds that with increasing productivity, the interconnected loading flow rate Q_{IC} is increasing, while the interconnected time is decreasing. Since there is a trade-off between the objectives, increasing the capacity utilization leads to the inverse behavior, namely longer loading times but smaller loading flow rates. The batch loading flow rate Q_B increases only very slightly with productivity (and hence decreases with increased capacity utilization) over the pareto front. A graph visualizing these behaviors has been included in the Supporting information, Fig. S2.

6 Concluding remarks

In this work we compared batch and CaptureSMB for a mAb capture using a protein A resin in terms of capacity utilization, that is how much product can be processed per resin volume and load/elute cycle, and productivity, that is how much product can be processed per resin volume and per time, at constant yield and purity. A dual loading flow rate strategy was used in both processes. A model for the description of both batch breakthrough experiments and continuous capture processes for protein A chromatography has been developed and verified. In order to fit the model, only a small number breakthrough experiments at different flow rates (and different concentrations, if desirable) in batch mode are needed, requiring only a small amount of feed. The resulting model has been validated for the CaptureSMB process and is a useful tool in predicting the performance in different operating points and optimizing the processes.

Using the model, a batch capture process and the twin-column CaptureSMB process have been optimized with respect to two different objectives, capacity utilization and productivity, with constant yield and purity. Due to the trade-off between the two objectives, pareto-fronts have been identified for both processes. Judging from these fronts, the CaptureSMB process shows an increase in productivity of around 40% at a given capacity utilization and an increase in capacity utilization of around 25% at a given productivity, compared to a batch process with the same total resin volume. In general, the CaptureSMB process shows better performance than the batch process, except if a very low productivity value (less than 10% of the maximum possible value) is admissible; in these cases the batch process shows slightly higher maximum capacity utilization than the CaptureSMB process.

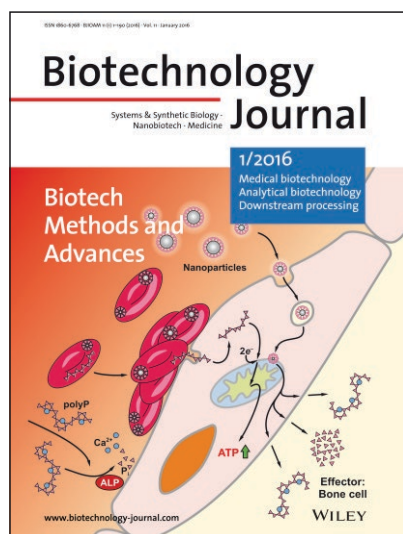
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The authors declare no financial or commercial conflict of interest.

7 References

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Cover illustration

This issue includes articles on an ultra-sensitive protein-protein interaction assay, an approach to overcome low yields of plant-made antibodies and a method to efficiently generate germline transgenic mice. The cover shows inorganic polyphosphates (polyP) that constitute an extracellular polymer for energy release, delivery and transport. PolyP use blood platelets as transport vehicles to reach their target cells, but have also been successfully encapsulated into microparticles allowing the development of novel therapeutics strategies. Image by Werner Müller.

Biotechnology Journal – list of articles published in the January 2016 issue.

Editorial

Biotechnology Journal – we are looking forward to a new decade
Jing Zhu

<http://dx.doi.org/10.1002/biot.201500668>

Review

Polyphosphate as a metabolic fuel in Metazoa: A foundational breakthrough invention for biomedical applications

Xiaohong Wang, Heinz C. Schröder and Werner E. G. Müller

<http://dx.doi.org/10.1002/biot.201500168>

Review

Quantum dot as probe for disease diagnosis and monitoring

Abhishek Mukherjee, Yumi Shim and Joon Myong Song

<http://dx.doi.org/10.1002/biot.201500219>

Review

Regulation of mesenchymal stem cell 3D microenvironment: From macro to microfluidic bioreactors

Sébastien Sart, Spiros N. Agathos, Yan Li and Teng Ma

<http://dx.doi.org/10.1002/biot.201500191>

Review

Potential applications of keratinocytes derived from human embryonic stem cells

Mohammad M. Movahednia, Fahad K. Kidwai, Doorgesh S. Jokhun, Christopher A. Squier, Wei Seong Toh and Tong Cao

<http://dx.doi.org/10.1002/biot.201500099>

Research Article

Selecting the optimal Tet-On system for doxycycline-inducible gene expression in transiently transfected and stably transduced mammalian cells

Atze T. Das, Xue Zhou, Stefan W. Metz, Monique A. Vink and Ben Berkhout

<http://dx.doi.org/10.1002/biot.201500236>

Research Article

Use of transposase and ends of IS608 enables precise and scarless genome modification for modulating gene expression and metabolic engineering applications in *Escherichia coli*

Chandresh Thakker, Kevin Lin, Heidi Martini-Stoica and George N. Bennett

<http://dx.doi.org/10.1002/biot.201500205>

Research Article

Ultra sensitive firefly luciferase-based protein-protein interaction assay (FlimPIA) attained by hinge region engineering and optimized reaction conditions

Makoto Kurihara, Yuki Ohmuro-Matsuyama, Keiichi Ayabe, Takahiro Yamashita, Hideki Yamaji and Hiroshi Ueda

<http://dx.doi.org/10.1002/biot.201500189>

Research Article

Monolith disk chromatography separates PEGylated protein positional isoforms within minutes at low pressure

Yu Isakari, Ales Podgornik, Noriko Yoshimoto and Shuichi Yamamoto

<http://dx.doi.org/10.1002/biot.201500294>

Research Article

Overcoming low yields of plant-made antibodies by a protein engineering approach

Julia Zischewski, Markus Sack and Rainer Fischer

<http://dx.doi.org/10.1002/biot.201500255>

Research Article

High yield of recombinant human apolipoprotein A-I expressed in *Pichia pastoris* by using mixed-mode chromatography

Vignesh N. Janakiraman, Abdelmajid Noubhani, Krishnan Venkataraman, Mookambeswaran Vijayalakshmi and Xavier Santarelli

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Research Article

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Research Article

Optimal model-based design of the twin-column CaptureSMB process improves capacity utilization and productivity in protein A affinity capture

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Rapid Communication

Asymmetric synthesis of aromatic β -amino acids using ω -transaminase: Optimizing the lipase concentration to obtain thermodynamically unstable β -keto acids

Sam Mathew, Seong-Su Jeong, Taeowan Chung, Sang-Hyeup Lee and Hyungdon Yun

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